Deep Learning

CS4112

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Training and Test Splits

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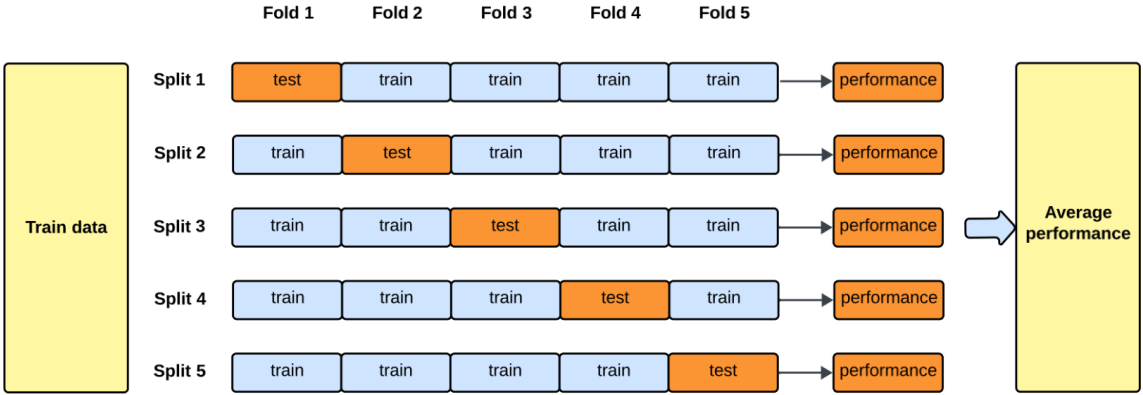
Training and Test Splits

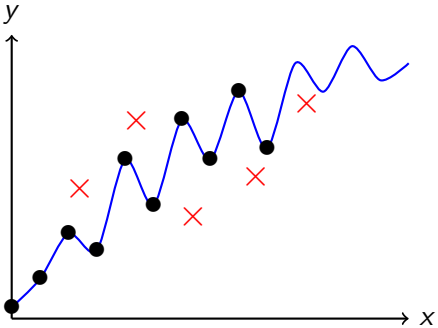
- Splits can be 2/3 for training and 1/3 for test
  - Must be random
  - Pandas has a DataFrame function 'sample' for this
- Can use K-Fold cross validation

K-fold Cross Validation



Training Error Vs Test Error

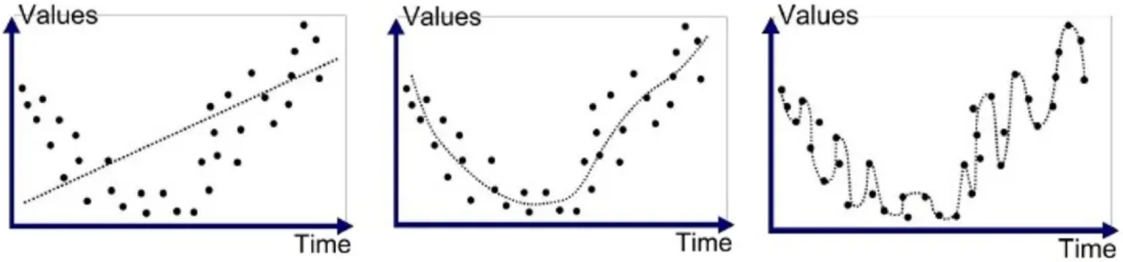
- Reducing Training Error is easy
- Test error is a better measure
- Important that test data is not touched during training



Training vs Test Error Pipeline

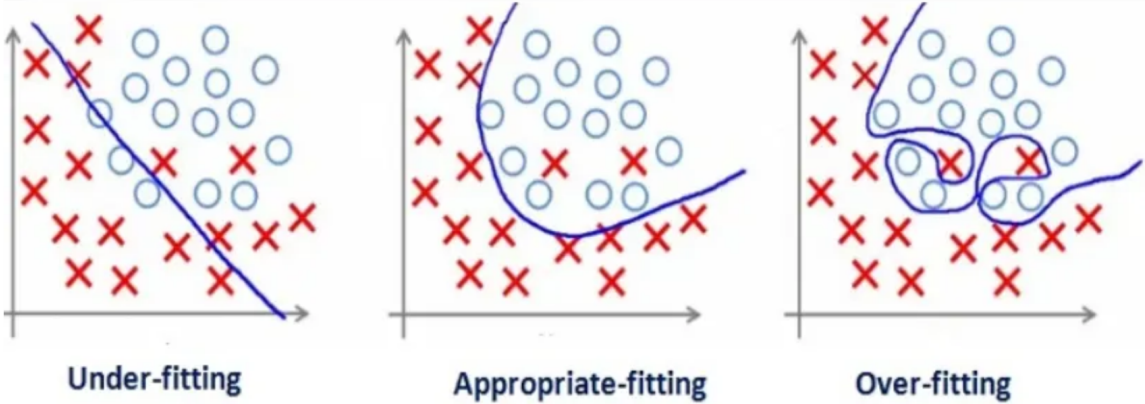
- Data which is known to us and the machine is the training data
- Data which is known to us only and not the machine is the test data
  - we can evaluation machine on the test data and fine-tune the hyper-parameters for better learning
- Data which is neither known to us nor the machine is the real world dataset
- **NOTE:** when we are confident enough that the machine has learned enough by evaluating its performance on the test data, then we can deploy the machine on the real world data.

Underfitting vs Overfitting in Polynomial Regression

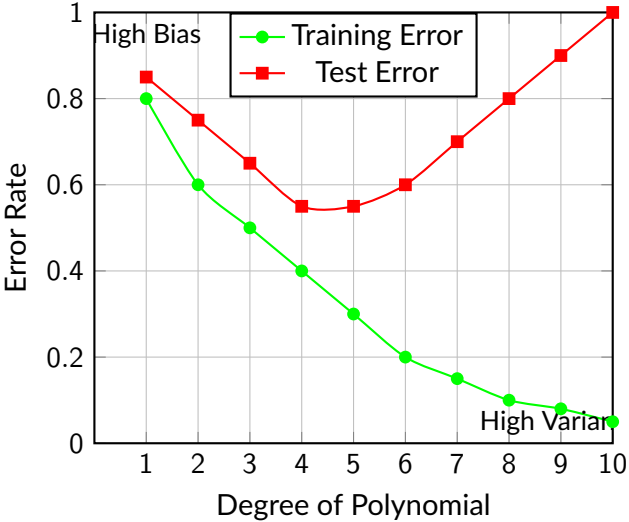


Underfitted | Good Fit/Robust | Overfitted

Underfitting vs Overfitting in Logistics Regression



Model Complexity



High Bias vs High Variance

High Bias

Assuming that the model is simpler than its actual complexity. High Bias is equivalent to underfitting.

High Variance

Assuming the model is much more complex than its actual complexity. High variance is equivalent to overfitting.

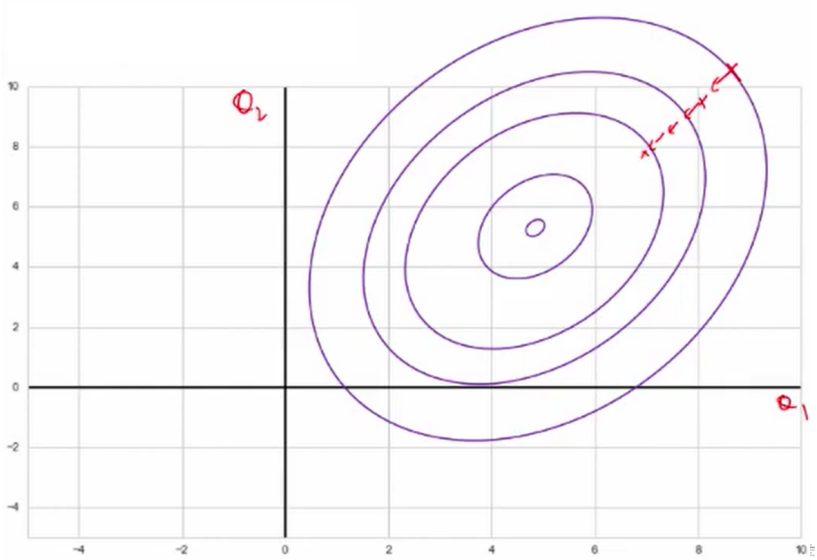
Balancing Parameters

$$w = 2z + az^2$$

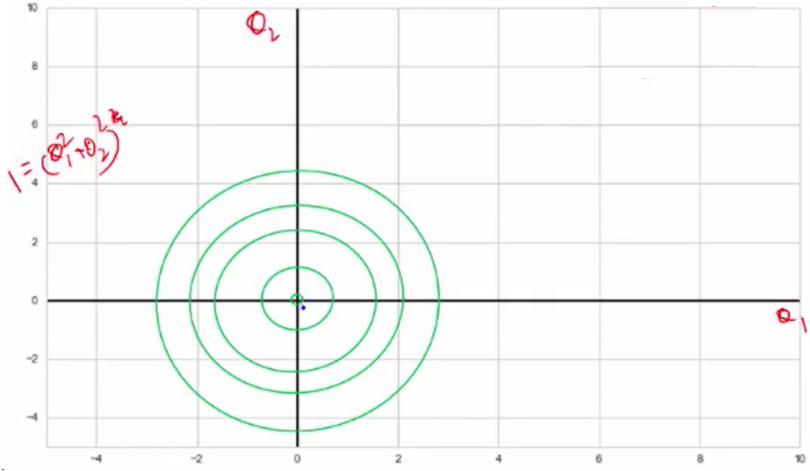
$$\begin{aligned} w &= 2(10) + 15(100) \\ &= 20 + 1500 \\ &= 1520 \end{aligned}$$

$$\begin{aligned} w &= 2(10) + 0.1(100) \\ &= 20 + 10 \\ &= 30 \end{aligned}$$

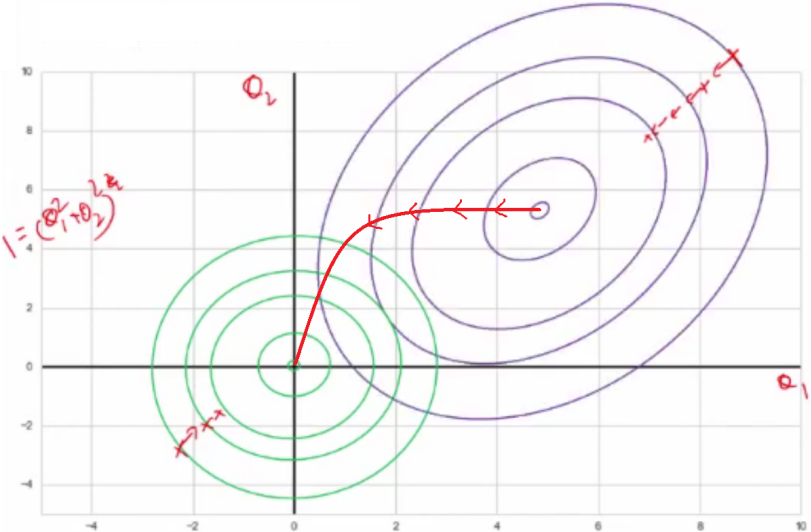
Fixing the Variance



Fixing the Variance



Fixing the Variance



Regularization

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2 + \lambda \sum_{i=1}^{n} \theta_i^2$$

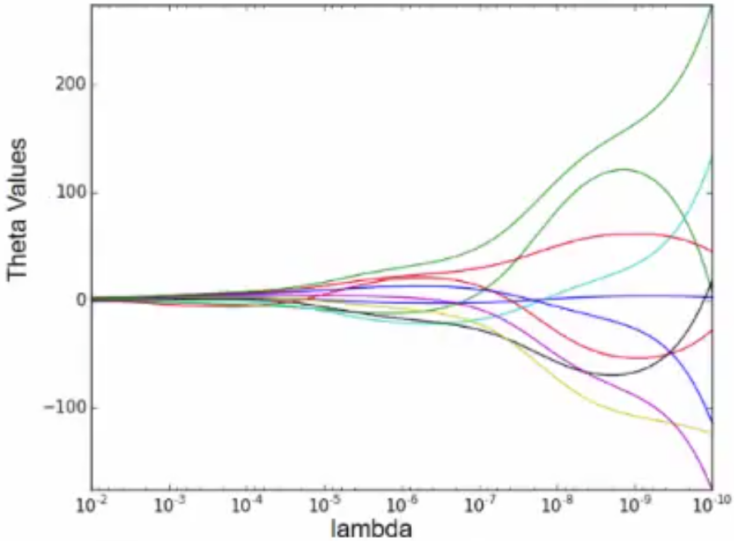
Regularization

Regularization

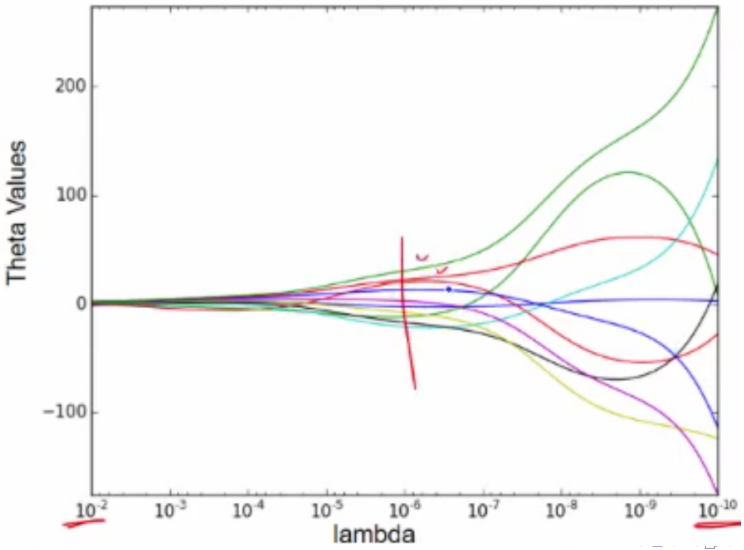
If the model has high variance (overfitting), regularization reduces it and brings the model to a balanced point with better generalization, i.e., just right

- reducing the complexity of an overfitted model to a point which is just right
- λ is called the Regularization Parameter
- Single parameter to adjust the complexity
- Since we have square of the model parameter, it's often called the L2-norm
- Alternative: L1-norm

Sparsity


Theta Values
200
100
0
−100
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
lambda

Sparsity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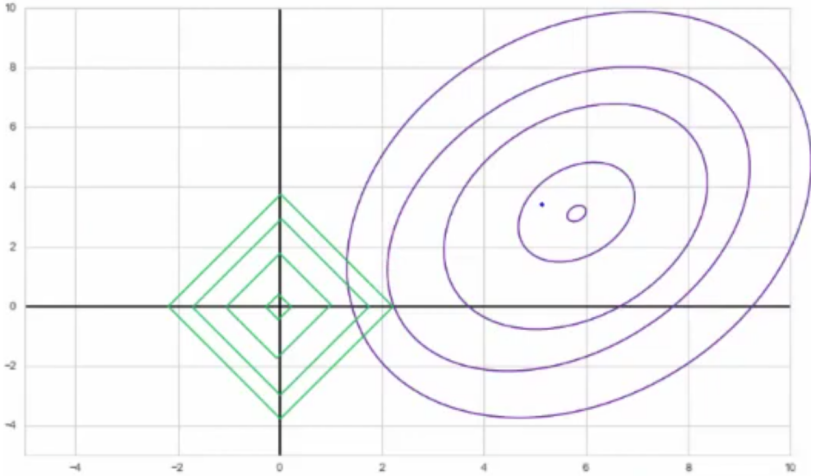
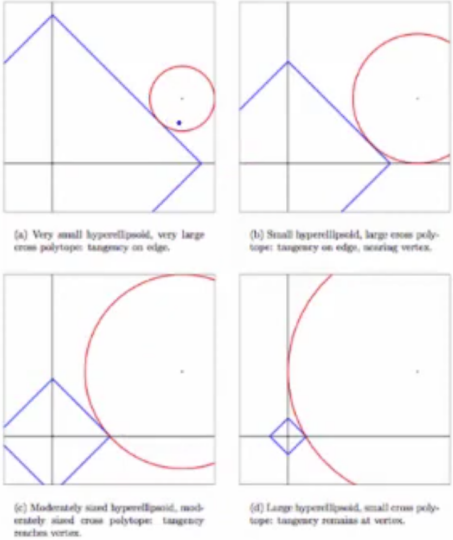
10^{-8}
10^{-9}
10^{-10}
lambda

L1-Norm



L1-Norm



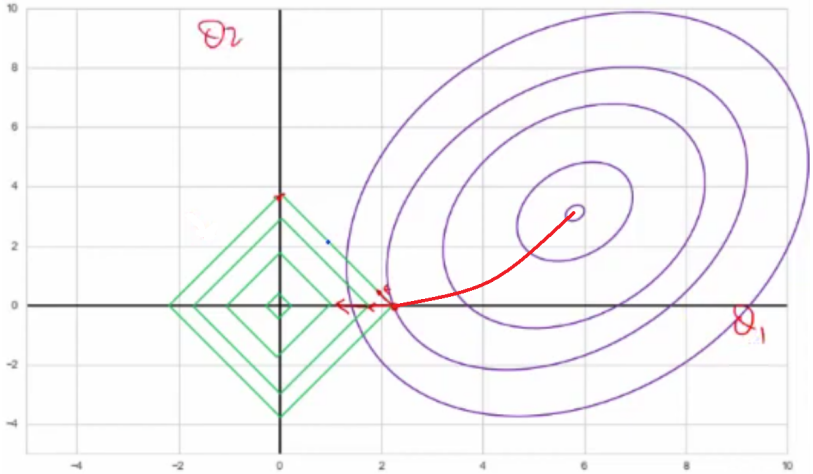
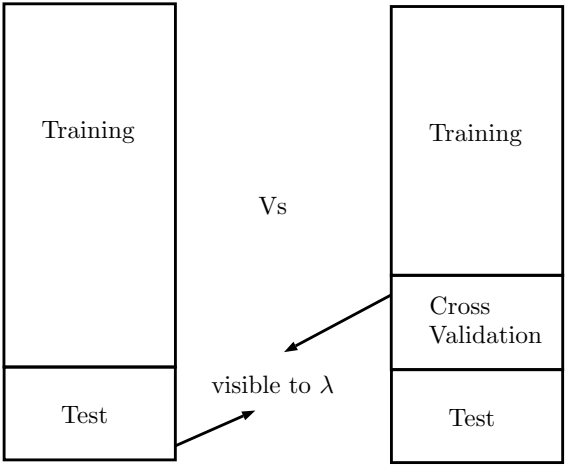
(a) Very small hyperellipsoid, very large cross polytope: tangency on edge.

(b) Small hyperellipsoid, large cross polytope: tangency on edge, nearing vertex.

(c) Moderately sized hyperellipsoid, moderately sized cross polytope: tangency reaches vertex.

(d) Large hyperellipsoid, small cross polytope: tangency remains at vertex.

L1-Norm



Cross Validation for Hyperparameters